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<body>

```

<h1>ðŸ©, The Circulatory System</h1>

<h2>Introduction</h2>

<div class="section">

1. Need for Nutrients, Gases and Liquids

All living organisms need nutrients, gases and liquids for their growth and maintenance. These are essential for keeping the body alive and functioning properly.

</div>

<div class="section">

2. Transport to All Body Parts

The required materials must be transported to all parts of the body. This transportation is necessary in both unicellular and multicellular organisms.

<div class="section">

3. Transport in Unicellular Organisms

In unicellular organisms, materials need to travel only short distances. Therefore, they do not require a complex transport system.

</div>

<div class="section">

4. Transport in Multicellular Organisms

In multicellular organisms, substances may have to travel very long distances. In the tallest plants, this distance may be around 100 feet.

</div>

<div class="section">

5. Transport in Amoeba and Hydra

Simple organisms like Amoeba and Hydra transport materials using simple processes – mainly diffusion and osmosis.

</div>

<div class="section">

6. Diffusion

Diffusion helps substances move without requiring a specialised circulatory system.

</div>

<div class="section">

7. Osmosis

Osmosis is another simple process involved in transportation. Organisms like Amoeba and Hydra use such simple processes to move materials.

</div>

<div class="section">

8. Problem in Large Multicellular Organisms

Large multicellular organisms contain trillions of cells. Using only diffusion and osmosis would take a very long time to transport materials.

</div>

<div class="section">

9. Development of a Specialised Transport System

To avoid this delay, organisms developed a separate specialised transport system that carries materials much faster and more efficiently throughout the body.

</div>

Circulatory System

Circulatory system is a specialised system that carries materials rapidly and efficiently throughout the body.

We:

-

- Eat solids

- Drink liquids

- Breathe gases

The circulatory system helps transport the materials required by different parts of the body.

Memory Trick
A-H D-O
Amoeba + Hydra Diffusion + Osmosis

Quiz

Quiz

-

- Which system carries materials efficiently throughout the body?

A) Digestive system B) Circulatory system C) Nervous system D) Skeletal system

- Amoeba transports materials mainly through:
A) Circulation only
B) Diffusion and osmosis C) Transpiration D) Blood vessels

- Why are diffusion and osmosis alone insufficient in large organisms?
A) Too fast B) Require blood C) Take a long time D) Stop respiration

- Living organisms require nutrients mainly for:
A) Growth and maintenance B) Hearing C) Movement only D) Reproduction only

- Humans eat solids, drink liquids and breathe:
A) Water B) Blood C) Gases D) Nutrients

3.1 Heart Beat

A doctor holds a patient's wrist and looks at a watch to count the heartbeat. Heartbeat can be indirectly counted through the pulse.

Activity

Activity-1 Observing Pulse

Procedure

-

- Place the index and middle fingers on the wrist below the thumb.

- A rhythmic pushing movement can be felt.

Pulse: the rhythmic up-and-down movement felt at the wrist.

- Count the pulse for one minute.
- Run at the same place for one minute.
- Again count the pulse for one minute.
- Take readings from at least two friends.

Table- 1

S.No.	Name of student	Pulse rate at rest	Pulse rate after jogging
1	â€	â€	â€
2	â€	â€	â€

Observation: Pulse rate is not the same at rest and after jogging. Physical activity changes pulse rate.

Fig-1: Pulse â€ shows placement of fingers on the wrist.

Activity-2 â€ Pulse Indicator

Pulse rate varies from person to person and from situation to situation.

When a person is:

- Afraid â€ pulse rate â€
- Excited â€ pulse rate â€
- Climbing stairs â€ pulse rate â€
- Running â€ pulse rate â€

There is a relationship between pulse rate and heartbeat.

Making a Pulse Indicator

- Take an injection bottle lid/shirt button.
- Insert a matchstick into it.
- Place it on the wrist.
- Observe the matchstick movements.
- Put the other hand on the chest and feel the heartbeat.
- Compare the movement of the pulse indicator with the heartbeat.

Fig-2: Pulse indicator.

<div class="fact">

ðŸ’jDo You Know? â€” Variation in Pulse Rate

Age/Group</th>	Pulse rate/min</th></tr>
Newborn, 0â€”3 months</td>	100â€”150</td></tr>
Infants, 3â€”6 months</td>	90â€”120</td></tr>
Infants, 6â€”12 months</td>	80â€”120</td></tr>
Children, 1â€”10 years</td>	70â€”130</td></tr>
Children over 10 & adults</td>	60â€”100</td></tr>
Well-trained adult athletes</td>	40â€”60</td></tr>

</div>

<div class="star">â€” Well-trained adult athletes may have a pulse rate as low as 40â€”60/min.</div>

<h3>Stethoscope â€” Rene Laennec</h3>

<p>Rene Laennec invented the stethoscope in 1816. Before its discovery, doctors listened to the heartbeat by placing their ear directly on the patient's chest. Laennec discovered that a paper tube helped him hear the heartbeat clearly. Later, he used bamboo instead of the paper tube. He called the instrument a stethoscope.</p>

<div class="mnemonic">ðŸŒŠ Laennec â†’ Listen (both start with L)</div>

<div class="star">â€” 1816 â†’ Rene Laennec â†’ Stethoscope</div>

<div class="activity">

<h4>Activity-3 â€” Make Your Own Stethoscope</h4>

Make a paper tube â€” Length: 10 inches, Diameter: 1 inch.

Keep one end near your ear.

Keep the other end on your friend's chest.

Listen to the heartbeat and count for one minute.

Count the friend's pulse rate.

Record observations of at least 10 students.

S.No.</th>	Name</th>	Heartbeat at rest/min</th>	Pulse rate at rest/min</th></tr>
1</td>	Eshwar</td>	72</td>	72</td></tr>

</div>

<div class="star">â€” Conclusion: Pulse rate = Heartbeat rate. The beat of the heart sends blood rushing to different parts of the body.</div>

Heart – Basic Facts

Heart is located between the lungs.

It is protected by the rib cage.

Its approximate size is equal to the size of your fist.

<div class="quiz">

– Quiz – Heart Beat

The rhythmic movement felt at the wrist is called:
A) Pressure B) Pulse C) Systole D) Diastole

Who invented the stethoscope?
A) William Harvey B) Marcello Malpighi C) Rene Laennec D) Fabrici

The stethoscope was invented in:
A) 1574 B) 1650 C) 1816 D) 1900

The heart is approximately the size of:
A) Head B) Fist C) Foot D) Kidney

Which group has a pulse rate of about 40–60/min?
A) Newborns B) Infants C) Children D) Well-trained adult athletes

</div>

<hr>

Lab Activity – Internal Structure of Mammalian Heart

<div class="activity">

Aim

<p>To observe the internal structure of the mammalian heart.</p>

Materials Required

Fresh sheep/goat heart

Soda straws

Used pen refills

Sharp long blade/scalpel

Dissection tray

Jug of water

Dissection scissors

Forceps

<p>The structure of mammalian hearts is similar; therefore a sheep or goat heart can be used.</p>

Procedure

<h3>Pericardium</h3>

<p>Heart is covered by a two-layered membrane called the pericardium/ pericardial membrane. The space between the layers contains pericardial fluid, which protects the heart from shocks.</p>

<h3>Chambers</h3>

<p>Heart has 4 chambers:</p>

<table>

<tr><th>Upper chambers</th><th>Lower chambers</th></tr>

<tr><td>Right atrium</td><td>Right ventricle</td></tr>

<tr><td>Left atrium</td><td>Left ventricle</td></tr>

</table>

Upper chambers â†’ atria/auricles

Lower chambers â†’ ventricles

Ventricular walls are thicker than atrial walls.

<h3>Coronary Vessels</h3>

<p>Coronary vessels â†’ blood vessels in the walls of the heart that supply blood to the heart muscles.</p>

<h2>Blood Vessels Attached to Heart</h2>

<h3>Arteries</h3>

Originate from the heart.

Supply blood to different organs.

Have thick walls.

Largest artery â†’ Aorta

Pulmonary artery carries blood from heart â†’ lungs.

<h3>Veins</h3>

Bring blood from body parts â†’ heart.

Have relatively thin walls.

<p>Superior Vena Cava â†’ Collects blood from anterior parts such as head and neck.</p>

<p>Inferior Vena Cava â†’ Brings blood from posterior body parts such as hands and legs.</p>

<p>Septa â†’ muscular partitions separating the chambers.</p>

<p>Valves â€™ openings between atria and ventricles are guarded by valves.</p>

<h2>Heart Valves</h2>

<div class="flow">Right atrium â€™ Tricuspid valve â€™ Right ventricle</div>

<div class="flow">Left atrium â€™ Bicuspid/Mitral valve â€™ Left ventricle</div>

<p>Pulmonary valves â€™ present in the region of the pulmonary aorta.

Systemic valves â€™ present in the region of the systemic aorta.</p>

<div class="flow">Body parts â€™ Superior & inferior vena cava â€™ Right atrium (deoxygenated blood)</div>

<div class="flow">Lungs â€™ Pulmonary veins â€™ Left atrium (oxygenated blood)</div>

<div class="flow">Left ventricle â€™ Aorta â€™ Body (oxygenated blood)</div>

<div class="flow">Right ventricle â€™ Pulmonary artery â€™ Lungs (deoxygenated blood)</div>

<div class="star">

â€™ Exam Points

Heart has 4 chambers.

Tricuspid â€™ right side.

Bicuspid/Mitral â€™ left side.

Aorta â€™ largest artery.

Pulmonary artery carries deoxygenated blood.

Pulmonary veins carry oxygenated blood.

Pericardial fluid protects the heart from shocks.

</div>

<div class="mnemonic">ðŸ§ TRI = RIGHT (TRicuspid â€™ Right) | BI = LEFT (Bicuspid â€™ Left)</div>

<div class="quiz">

<h4>â€™ Quiz</h4>

The membrane surrounding the heart is:
A) Pleura B) Pericardium C) Septum D) Endothelium

Pericardial fluid protects the heart from:
A) Digestion B) Shocks C) Oxygen D) Blood

The valve between right atrium and right ventricle is:
A) Bicuspid

B) Mitral C) Tricuspid D) Systemic

Which vessel brings oxygenated blood from lungs?
A) Pulmonary artery B) Pulmonary veins C) Vena cava D) Aorta

Coronary vessels supply blood to:
A) Lungs B) Brain C) Heart muscles D) Kidneys

</div>

<hr>

<h2>3.3 Blood Vessels and Circulation</h2>

<h3>Girolamo Fabrici</h3>

<p>In 1574, Italian doctor Girolamo Fabrici studied veins in the leg. He noticed small valves in veins that allowed blood to flow towards the heart – one-way valves that prevented backflow. Movement of leg muscles helps blood move through veins. Fabrici did not recognise the full importance of his discovery.</p>

<h3>William Harvey</h3>

<p>William Harvey (1578–1657) was an English doctor. He studied under Fabrici in Italy, dissected human hearts, and studied valves between atria and ventricles. He recognised that they were one-way valves.</p>

<div class="flow">Atrium – Valve – Ventricle</div>

<p>During contraction: Ventricle – Artery. The blood cannot flow backwards into the atrium.</p>

<h3>Harvey's Experiments</h3>

<p>Harvey blocked different veins in animals. Veins bulged on the side of the block away from the heart, showing blood in veins was trying to move towards the heart. In arteries, blood bulged on the heart side of the block, showing arterial blood was moving away from the heart.</p>

<h3>Harvey's Conclusion</h3>

<div class="flow">Heart – Arteries – Body/Lungs – Veins – Heart</div>

<p>Blood therefore circulates.</p>

<div class="fact">

Pulmonary Circulation: Right ventricle – Lungs – Left atrium

Systemic Circulation: Left ventricle – Aorta – Body – Veins – Right atrium

</div>

Who observed valves in leg veins in 1574?
A) Harvey B) Fabrici C) Malpighi D) Laennec

William Harvey lived during:
A) 1578â€“1657 B) 1628â€“1694 C) 1700â€“1780 D) 1816â€“1880

Who discovered the connection between tiny arteries and veins?
A) Fabrici B) Harvey C) Malpighi D) Laennec

The word capillary is related to the Latin word for:
A) Blood B) Hair C) Heart D) Tube

Blood in veins normally moves:
A) Away from heart B) Towards heart C) Only towards lungs D) Only towards kidneys

</div>

<hr>

<h2>3.4 Arteries and Veins</h2>

<p>Arteries â†’ heart to body parts.
Veins â†’ body organs to heart.</p>

<h3>3.4.1 Blood Capillaries</h3>

Capillaries are microscopic blood vessels.

Their walls consist of a single layer of cells.

They allow diffusion of substances.

Leucocytes/WBCs can squeeze through the capillary wall.

Capillaries establish continuity between arteries and veins.

<div class="fact">

Fig-9 Labels

Artery: Elastic fibrous coat, Muscle layer, Endothelium, Lumen

Vein: Tough fibrous coat, Endothelium, Lumen

Blood capillary: Very thin wall/single cell layer

</div>

<h3>Artery vs Vein</h3>

<table>

<tr><th>Feature</th><th>Artery</th><th>Vein</th></tr>

<tr><td>Walls</td><td>Thick</td><td>Thin</td></tr>

<tr><td>Valves</td><td>Absent</td><td>Present</td></tr>

<tr><td>Shape without blood</td><td>Can retain</td><td>Cannot retain as well</td></tr>

<tr><td>Blood direction</td><td>Heart â†’ organs</td><td>Organs â†’ heart</td></tr>

<tr><td>Pressure</td><td>High</td><td>Low</td></tr>

<tr><td>Usually carries</td><td>Oxygenated</td><td>Deoxygenated</td></tr>

Important exception	Pulmonary artery
Pulmonary vein	

Pulmonary artery carries deoxygenated blood.
 Pulmonary vein carries oxygenated blood.

Activity-4

Leg observation

- Sit with one leg dangling.
- Rest the back of the other knee on it.
- Small movements can be felt with each heartbeat.
- Keeping this position too long reduces blood flow – may produce pins and needles.

Vein-valve observation

- Swing the arm several times to fill veins with blood.
- Hold arm vertically downward.
- Press along a prominent vein in the direction opposite to normal blood flow – towards the hand.
- Swellings can be observed where blood is pushed against valves.

Think and Discuss:

- Why are artery walls strong and elastic?
- Why can arteries be compared to a tree dividing into smaller branches?
- Why is the lumen of a vein larger than that of an artery?

A = Away (Artery = Away from heart)
 V = Visit (Veins Visit the heart)

Quiz

- Capillaries are made of:
 - A) Three layers
 - B) Two layers
 - C) Single layer of cells
 - D) Muscle only
- WBCs can squeeze through:
 - A) Heart wall
 - B) Capillary wall
 - C) Aorta
 - D) Valve
- Arteries carry blood:
 - A) Towards heart
 - B) Away from heart
 - C) Only

to lungs D) Only to brain

Which normally contains valves?
A) Arteries B) Veins C) Aorta only D) Capillaries only

The pulmonary artery carries:
A) Oxygenated blood B) Deoxygenated blood C) Lymph D) Tissue fluid

</div>

3.5 The Cardiac Cycle

<p>Human heart begins beating around the 21st day of embryonic development. It beats rhythmically until death. If the heart stops beating, the person dies.</p>

Cardiac Cycle: One contraction + one relaxation of atria and ventricles = one cardiac cycle.

Sequence

<0|>

Diastole " All four chambers are relaxed.

Filling of atria â€” Vena cava â†’ Right atrium; Pulmonary veins
â†’ Left atrium

Atrial contraction → Atria contract → blood enters ventricles.

Ventricular contraction – Ventricles contract; atrioventricular valves close; first sharp heart sound – **LUB**; blood moves into Aorta / Pulmonary artery.

Ventricular relaxation â€” Pressure falls; blood in aorta tries to return; semilunar valves close, preventing backflow into ventricles; second heart sound â†’ DUB; atria fill again; cycle repeats.

Systole and Diastole

Systole â†’ active/contraction phase.
Diastole â†’ resting/relaxation phase.

| Event | Time |

Atrial systole	0.11–0.14 sec
----------------	---------------

| Ventricular systole | 0.27±0.35 sec |

<tr><td>Joint diastole</td><td> ∞ 0.4 sec</td></tr>
--

Complete cardiac cycle	≈ 0.8 sec
------------------------	-------------------

[illegible]

<p>Blood passes through the heart twice to complete one circulation.</p>

<table>	
<tr><th>Single Circulation</th><th>Double Circulation</th></tr>	
<tr><td>Heart crossed once</td><td>Heart crossed twice</td></tr>	
<tr><td>Diagram includes gills</td><td>Diagram includes lungs</td></tr>	
<tr><td>One circulation pathway</td><td>Pulmonary + systemic pathways</td></tr>	
</table>	

<div class="flow">Human Double Circulation:
Right ventricle â†’ Lungs
â†’ Left atrium â†’ Left ventricle â†’ Body â†’ Right atrium â†’ Right
ventricle</div>

<div class="mnemonic">ðŸ§ Single = Single heart visit | Double = Double heart visit</div>

<div class="quiz">
<h4>â “ Quiz</h4>

In single circulation, blood passes through the heart:
A) Once B) Twice C) Three times D) Four times
In double circulation, blood passes through the heart:
A) Once B) Twice C) Three times D) Never
Which respiratory organ appears in the textbook single-circulation figure?
A) Lungs B) Gills C) Skin D) Trachea
Human circulation is:
A) Single B) Double C) No circulation D) Diffusion only
Which circulation involves lungs?
A) Pulmonary circulation B) Root circulation C) Lymph only D) Single circulation only

</div>

<hr>

<h2>3.7 Lymphatic System</h2>

<p>After sitting without movement during a long overnight journey, footwear may feel tight. In older people, the lower legs may visibly swell. This condition is called oedema.</p>

<h3>Why does swelling occur?</h3>

<p>As blood passes through capillaries, some fluid and certain materials move out into tissues. These materials must be collected and returned to blood circulation.</p>

<h3>Tissue Fluid</h3>

<div class="flow">Heart â†’ Arteries â†’ Capillaries â†’ Liquid containing nutrients moves into tissues</div>

<p>This liquid is called tissue fluid.</p>

<h3>Lymphatic System</h3>

<p>Tissue fluid flows through the lymphatic system. Most of it ultimately returns to the bloodstream.</p>

<h3>Lymph</h3>

<p>Lymph forms a vital link between blood and tissues.</p>

</body>

</html>